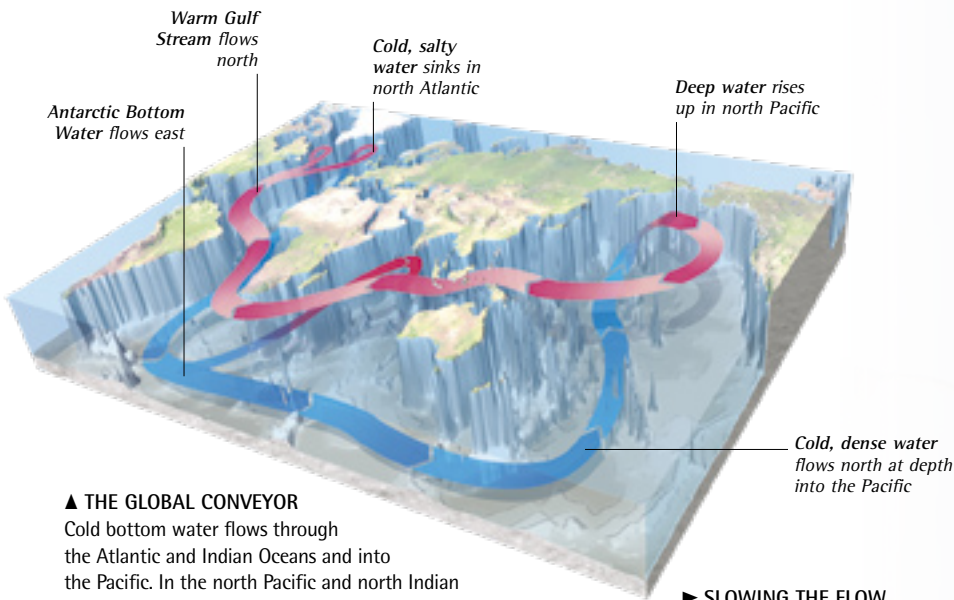




▲ THE GLOBAL CONVEYOR

Cold bottom water flows through the Atlantic and Indian Oceans and into the Pacific. In the north Pacific and north Indian Oceans, some of it feeds into the surface current gyres. These link with surface currents in the Atlantic, where the Gulf Stream supplies the water that sinks in the far north to help drive the flow. This is often known as the global conveyor, because it conveys ocean water all around the globe.

► SLOWING THE FLOW

Global warming could weaken deep-water currents by reducing the amount of water sinking in cold oceans. This is partly because there is less ice forming on the ocean surface. But fresh water draining off the melting polar ice sheets into the sea also makes seawater less salty and dense, and less likely to sink. This is already reducing the flow of the North Atlantic Deep Water, which is the driving force behind the ocean conveyor.



▲ VITAL SUPPLIES

A vast amount of heat is moved around the world by the ocean conveyor. This helps prevent the cold polar regions from becoming even colder, and stops the tropics from becoming so hot that life there would be impossible. The deep-water circulation is also essential to marine life like these jellyfish, because it carries oxygen down to the ocean floors and draws vital nutrients up to the surface.



▲ CLIMATE CHAOS

If the global conveyor is weakened by climate change, it will affect the whole world, but the effect on northern Europe could be catastrophic. Sinking water near Iceland draws the warm Gulf Stream north, giving Europe its mild climate. If the Gulf Stream were to fail, European cities could suffer Arctic winters.

